
CHICAGO JOURNALS

Magnetostratigraphy across the Wasatchian/Bridgerian NALMA Boundary (Early to Middle Eocene) in the Western Green River Basin, Wyoming
Author(s): William C. Clyde, John-Paul Zonneveld, John Stamatakos, Gregg F. Gunnell, and William S. Bartels
Source: *The Journal of Geology*, Vol. 105, No. 6 (November 1997), pp. 657-670
Published by: The University of Chicago Press
Stable URL: http://www.jstor.org/stable/10.1086/515970
Accessed: 12/12/2015 08:41

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at
http://www.jstor.org/page/info/about/policies/terms.jsp

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.


The University of Chicago Press is collaborating with JSTOR to digitize, preserve and extend access to *The Journal of Geology*.

http://www.jstor.org

This content downloaded from 23.235.32.0 on Sat, 12 Dec 2015 08:41:09 AM
All use subject to JSTOR Terms and Conditions

ARTICLES

Magnetostratigraphy across the Wasatchian/Bridgerian NALMA Boundary (Early to Middle Eocene) in the Western Green River Basin, Wyoming[1]

William C. Clyde, John-Paul Zonneveld,[2] John Stamatakos,[3] Gregg F. Gunnell, and William S. Bartels[4]

Museum of Paleontology, The University of Michigan, Ann Arbor, MI 48109-1079

ABSTRACT

New paleomagnetic and biostratigraphic data from the western Green River Basin of Wyoming provide a magnetobiostratigraphic framework to correlate the late Wasatchian through early Bridgerian North American Land Mammal Ages (NALMA) to the Geomagnetic Polarity Time Scale (GPTS). Three-hundred and twelve paleomagnetic samples were collected from 73 sites within a 350-m continuously exposed flat-lying section. The section is characterized by alternating packages of well-developed paleosol mudstone horizons (Wasatch Formation) and lacustrine shale/limestone horizons (Green River Formation). Fossil mammal assemblages are known from various levels within the section and constrain the location of the Wasatchian/Bridgerian NALMA boundary to lie between 60 and 120 m. Paleomagnetic results of paleosol horizons are generally characterized by two components, a present-day field component with low unblocking temperatures up to 400°C, and a characteristic component with unblocking temperatures up to 680°C. Characteristic magnetizations have either N-NW and steeply down directions or antipodal S-SE and up directions. The characteristic component is interpreted as primary based on antipodal polarity directions and congruence with the expected Eocene direction for western Wyoming. Results indicate that the first 30 m of section are characterized by normal polarity, the next 230 m by dominantly reversed polarity, followed by a 70 m interval of alternating polarity. Given radiometric ages for the late Wasatchian (early Lostcabinian ''subage'') and middle Bridgerian, as well as previous Wasatchian and Bridgerian magnetostratigraphic results, two potential correlations to the GPTS are possible. Correlation 1 places the Wasatchian/Bridgerian NALMA boundary within Chron C23r at about 52 Ma. Correlation 2 places the boundary within Chron C22r at about 50.5 Ma. Correlation 2 is tentatively preferred as it agrees most closely with previous estimates for the age of the boundary and assumes more uniform sediment accumulation rates.

Introduction

Correlation of the North American Land Mammal Age (NALMA) framework to the Geomagnetic Polarity Time Scale (GPTS) represents an important step toward creating a globally integrated stratigraphic framework (Berggren et al. 1985, 1995). The Wasatchian/Bridgerian NALMA boundary is characterized by significant biotic turnover and broadly corresponds to the global temperature maximum for the entire Cenozoic, making it an important period in the North American terrestrial record (Stucky 1984; Krishtalka et al. 1987; Gunnell and Bartels 1994; Gunnell 1997 unpub. data). No previous studies have analyzed the magnetostratigraphy through this boundary, leaving this part of the NALMA framework poorly constrained in time (Prothero 1995). Magnetostratigraphic studies from the Bighorn Basin correlated early to late Wasatchian strata to Chron C24 (Butler et al. 1981; Clyde et al. 1994; Tauxe et al. 1994). Flynn (1986) correlated early Bridgerian strata from the East Fork Basin to Chron C21 or C22 (see also McCarroll et al. 1996 and Walsh 1996). On the basis of these correlations, the Wasatchian/Bridgerian NALMA boundary is loosely constrained to fall somewhere in the 5 myr interval between Chron C24n and Chron C21n. In order to refine further the timing of this boundary, we analyzed the magnetostratigraphy of a 350-m section of alternating alluvial (Wasatch Formation) and lacustrine (Green River

[1] Manuscript received January 14, 1997; accepted June 10, 1997.

[2] Department of Earth and Atmospheric Sciences, University of Alberta, I-26 Earth Sciences Building, Edmonton, Alberta, T6G 2E3, Canada.

[3] Center for Nuclear Waste Regulatory Analysis, Southwest Research Institute, 6220 Culebra Rd., San Antonio, TX, 78238.

[4] Department of Geological Sciences, Albion College, Albion, MI, 49224.

[The Journal of Geology, 1997, volume 105, p. 657–669] © 1997 by The University of Chicago. All rights reserved. 0022-1376/97/10506-0003$01.00

This content downloaded from 23.235.32.0 on Sat, 12 Dec 2015 08:41:09 AM
All use subject to JSTOR Terms and Conditions

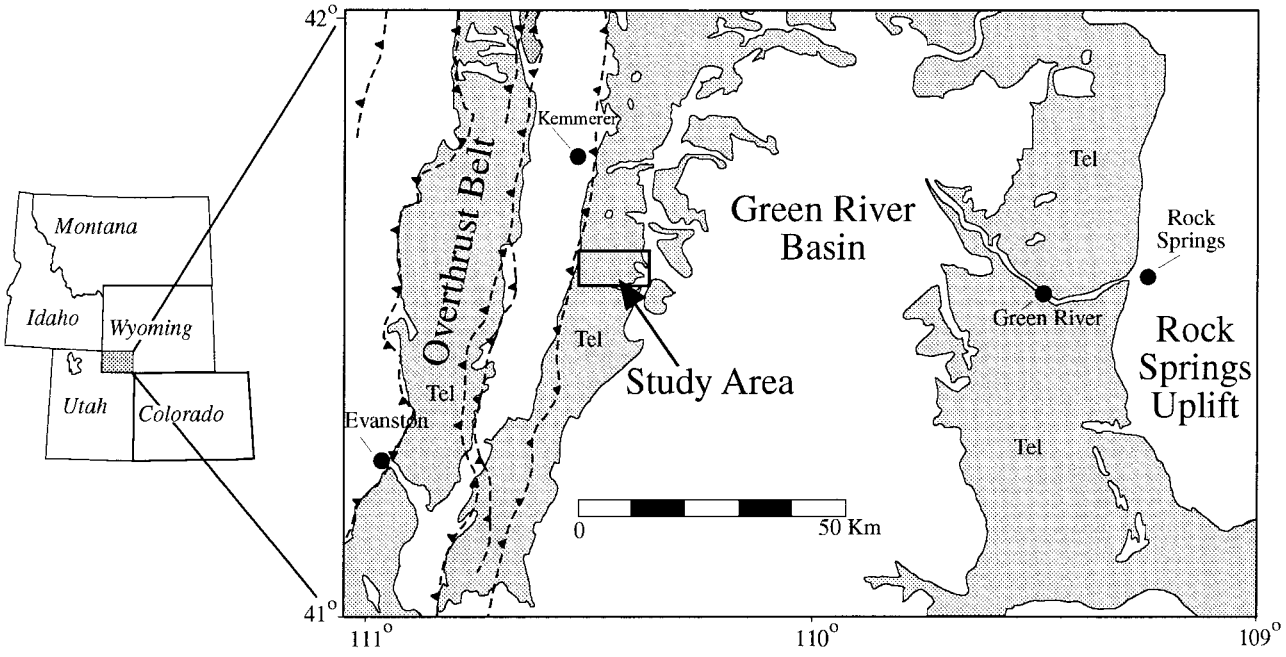


Figure 1. Location map showing the study area in the western Green River Basin of southwestern Wyoming. Shaded areas represent lower Eocene (Tel) deposits of the Wasatch Formation and Green River Formation. Adapted from Love and Christiansen (1985) and M'Gonigle and Dover (1992).

Formation) sediments from the western Green River Basin that preserves a well-documented mammalian succession across the Wasatchian/Bridgerian boundary (see figure 1 for location map).

The Wasatchian/Bridgerian NALMA Boundary

The majority of recent work on early and middle Eocene NALMA magnetobiochronology has been devoted to constraining the Clarkforkian/Wasatchian and Bridgerian/Uintan NALMA boundaries in time (Butler et al. 1981; Clyde et al. 1994; Tauxe et al. 1994; Prothero and Emry 1997). Relatively little attention has been devoted to the intervening Wasatchian/Bridgerian boundary, even though it represents a critical period in the climatic and biotic evolution of North America.

In many ways, the Wasatchian/Bridgerian boundary interval represents a climatic turning point in North America. Paleoclimatic indicators show mean annual temperatures (MAT) in North America increasing gradually through the early Eocene, reaching a Cenozoic maximum during late Wasatchian and early Bridgerian time (Wing et al. 1991; Wing and Greenwood 1993). Additional evidence points to increased equability and dramatically reduced latitudinal temperature gradients during this period as well, suggesting a global climate very different from that of today (Hutchison 1982; Wing and Greenwood 1993; Zachos et al. 1994; Markwick 1994). Floral compositions, mammalian body-size distributions, and increased mammalian arborealism in both Europe and North America indicate that these continents were at their most tropical during this interval (MacGinitie et al. 1974; Collinson and Hooker 1987; Wing and Greenwood 1993; Gunnell 1997). After these "greenhouse" conditions reached their apex near the Wasatchian/Bridgerian boundary, a directed and long-lasting global cooling took over, driving climates toward the "ice-house" conditions of the late Eocene and early Oligocene (Zachos et al. 1994).

A variety of important biotic changes occurred during this North American climate extreme. It was a period of generally increased endemism in North American mammalian faunas, punctuated by small, yet significant, immigration events (Woodburne 1997). The first appearance of several perissodactyls (e.g., *Hyrachyus, Palaeosyops*) and primates (e.g., *Omomys, Smilodectes*) marks the beginning of the Bridgerian NALMA and represents the most important of these immigrations (Krishtalka et al. 1987). One of the striking effects of this immigration was the transition from anaptomorphine-dominated to omomyine-dominated tarsiiform primate faunas (Gunnell 1997). Reptilian diversity reached a maximum during this period as

This content downloaded from 23.235.32.0 on Sat, 12 Dec 2015 08:41:09 AM
All use subject to JSTOR Terms and Conditions

well, with abundant aquatic turtles and crocodilians indicating warm and wet environmental conditions (Hutchison 1982). In sum, the extreme climate of the late Wasatchian and early Bridgerian had an important impact on terrestrial ecosystems.

In order to correlate and compare changes in the terrestrial record at this time to coincident changes in the marine realm, it is necessary to develop a robust chronological framework for relevant continental deposits. The section studied here from the western Green River basin represents one of the most complete and best-documented continental sequences across the Wasatchian/Bridgerian boundary making it an ideal place to develop an integrated magnetobiochronology for this part of the NALMA framework.

Stratigraphic Setting

There are two distinct lithologies present in the study area: fine-grained red/gray paleosol mudstone packages, typical of the Wasatch Formation, and fine-grained greenish gray shales interbedded with thinly bedded limestones, typical of the Green River Formation. The Wasatch Formation is interpreted as pedogenically altered fluvial overbank deposits. The Green River Formation is interpreted as unaltered passive accumulation within a lacustrine depositional system. Wasatch and Green River lithologies intertongue extensively throughout the western Green River Basin, tracking changes in the size and shape of paleolake Gosuite. The 350-m section studied here preserves several alternations between Wasatch and Green River deposition.

Thirty-three fossil localities correlated to this section provide a well-constrained biostratigraphic framework. Most localities are in Wasatch Formation paleosols and probably sample autochthonous accumulation of material on the Eocene floodplain. Localities from the lowest 60 m preserve typical late Wasatchian (Lostcabinian "subage") taxa (e.g., *Lambdotherium, Cantius*), whereas localities from the upper 230 m of section preserve typical Bridgerian taxa (e.g., *Hyrachyus, Notharctus*). The Wasatchian/Bridgerian NALMA boundary thus lies somewhere in a poorly fossiliferous interval between 60 and 120 m (see Krishtalka et al. 1987 for general discussion of Wasatchian and Bridgerian biostratigraphy). Taxonomic lists for Wasatchian and Bridgerian localities correlated to this section are given in tables 1 and 2. A complete stratigraphic and taxonomic accounting of these localities can be found in Zonneveld (1994). The Gardnerbuttean "subage" (Robinson 1966) is here considered part of the Bridgerian NALMA (following Stucky 1984; Krishtalka et al. 1987; and Gunnell and Bartels 1994).

Sampling Procedure and Sample Behavior

Three to nine oriented hand samples were collected from 73 sites (312 total samples) through the 350-m section. Average spacing between sites was 5.5 m but varied between 0.15 m and 24.7 m, depending on the quality of exposures. Samples were taken from one of three lithologies: (1) red B horizons of Wasatch Formation paleosols, (2) grayish green shales of the Green River Formation, and (3) grayish tan limestones of the Green River Formation. Weathered material was removed from the surface before extracting samples. In some cases this involved excavating pits up to 1 m into the outcrop. Samples were either cut into approximately 8 cm^3 cubes or cored into 1 inch diameter cylinders for paleomagnetic analysis.

Bulk susceptibility of samples was measured on a kappa bridge before demagnetization. Progressive thermal demagnetization and alternating field (af) demagnetization were performed using a Schonstedt TSD-1 thermal demagnetizer and a SI-4 af demagnetizer. Remanence measurements were made on a three-axis cryogenic magnetometer (2G) in the magnetically shielded paleomagnetic laboratory at the University of Michigan. Pilot studies showed that thermal demagnetization proved to be more effective in isolating remanence components, and thus nearly all samples were thermally demagnetized (10–20 steps) up to a maximum temperature of 680°C. Overprint and remanence components were determined by visual inspection of vector endpoint diagrams. Component directions were computed using least-squares analysis (Kirschvink 1980). Mean directions and their surrounding statistical distributions were calculated following the method described by Fisher (1953).

Magnetic Results

Samples from paleosol horizons were characterized by relatively strong NRM intensities (typically >1 mA/M). Thermal demagnetization provided stable demagnetization behavior in 118 of the 148 paleosol samples that were analyzed (80%). The majority of stable samples exhibited two components of magnetization (figure 2). The first component is characterized by low unblocking temperatures up to 400°C, and N, NE, and steeply down directions. This low-unblocking temperature component is interpreted as a present-day overprint based on its single polarity, and congruence with the present-

This content downloaded from 23.235.32.0 on Sat, 12 Dec 2015 08:41:09 AM
All use subject to JSTOR Terms and Conditions

Table 1. Mammalian Faunal List for Wasatchian Localities (BB110, BB114, BB115, BB116, BB118, BB128, BB129)

Order Marsupialia
  Family Didelphidae
    Peratherium innominatum (BB118)
    Peratherium marsupium (BB110)
Order Proteutheria
  Family Pantolestidae
    Palaeosinopa cf. *P. lutreola* (BB110)
  Family Lepticidae
    Palaeictops bicuspis (BB110)
Order ?Primates
  Family Microsyopidae
    Microsyops cf. *M. scottianus* (BB114)
  Family Paromomyidae
    Phenacolemur cf. *P. jepseni* (BB110)
Order Primates
  Family Adapidae
    Cantius frugivorous (BB114, 116, 110, 129)
  Family Omomyidae
    Anemorhysis cf. *A. wortmani* (BB110)
Order Condylarthra
  Family Meniscotheriidae
    Meniscotherium chamense (BB114, 110)
  Family Hyopsodontidae
    Hyopsodus miticulus (BB110)
    Hyopsodus wortmani (BB114, 110)
Order Perissodactyla
  Family Equidae
    Hyracotherium vasacciense (BB114, 110)
  Family Brontotheriidae
    Lambdotherium popoagicum (BB114, 115)
Order Artiodactyla
  Family Dichobunidae
    Diacodexis secans (BB110)
    Bunophorus sp. (BB118)
Order Carnivora
  Family Miacidae
    Miacis latidens (BB110)
    Vulpavus profectus (BB110)
  Family Viverravidae
    Viverravus minutus (BB110)
Order Rodentia
  Family Paramyidae
    Paramys cf. *P. copei* (BB114)
  Family Sciuravidae
    Sciuravus nitidus (BB110)
    Knightomys depressus (BB110)
Order Incertae sedis
  Family Metacheiromyidae
    Palaeanodon sp. (BB110)

Note. Localities in parentheses after taxon name indicate localities from which taxon is known.

Table 2. Mammalian Faunal List for Bridgerian Localities (BB012–14, BB019, BB035–38, BB044, BB051, BB068–70, BB074–77, BB096–98, BB103–104, BB109, BB117, BB122)

Order Marsupialia
  Family Didelphidae
    Peratherium innominatum (BB109)
    Peratherium marsupium (BB037)
    Peradectes chesteri (BB037)
Order Proteutheria
  Family Apatemyidae
    Apatemys bellulus (BB037)
Order Insectivora
  Family Dormaalidae
    Scenopagus curtidens (BB037)
Order ?Primates
  Family Microsyopidae
    Microsyops cf. *M. elegans* (BB122, 013, 036, 074)
Order Primates
  Family Adapidae
    Notharctus robinsoni (BB109, 037, 044)
    Smilodectes mcgrewi (BB074, 037)
  Family Omomyidae
    Omomys carteri (BB070, 074)
    Washakius insignis (BB037)
Order Mesonychia
  Family Mesonychidae
    Mesonyx cf. *M. obtusidens* (BB117)
Order Condylarthra
  Family Hyopsodontidae
    Hyopsodus minusculus (BB013)
Order Perissodactyla
  Family Equidae
    Orohippus cf. *O. pumilus* (BB013, 075, 019)
  Family Brontotheriidae
    Palaeosyops fontinalis (BB096, 122, 012)
  Family Helaletidae
    Hyrachyus sp. (BB117, 103, 074, 077, 037, 044)
    Gen et sp. nov. (BB103)
Order Carnivora
  Family Viverravidae
    Viverravus minutus (BB122)
Order Creodonta
  Family Hyaenodontidae
    ?*Prolimnocyon* sp. (BB122)
Order Rodentia
  Family Sciuravidae
    Sciuravus nitidus (BB013, 074, 109, 037)
  Family Paramyidae
    Leptotomus parvus (BB013, 103)
    Paramys cf. *P. copei* (BB013)
Order Incertae sedis
  Family Metacheiromyidae
    Metacheiromys sp. (BB074, 037)
  Family Epoicotheriidae
    Tetrapassalus sp. (BB109)

Note. Localities in parentheses after taxon name indicate localities from which taxon is known.

This content downloaded from 23.235.32.0 on Sat, 12 Dec 2015 08:41:09 AM
All use subject to JSTOR Terms and Conditions

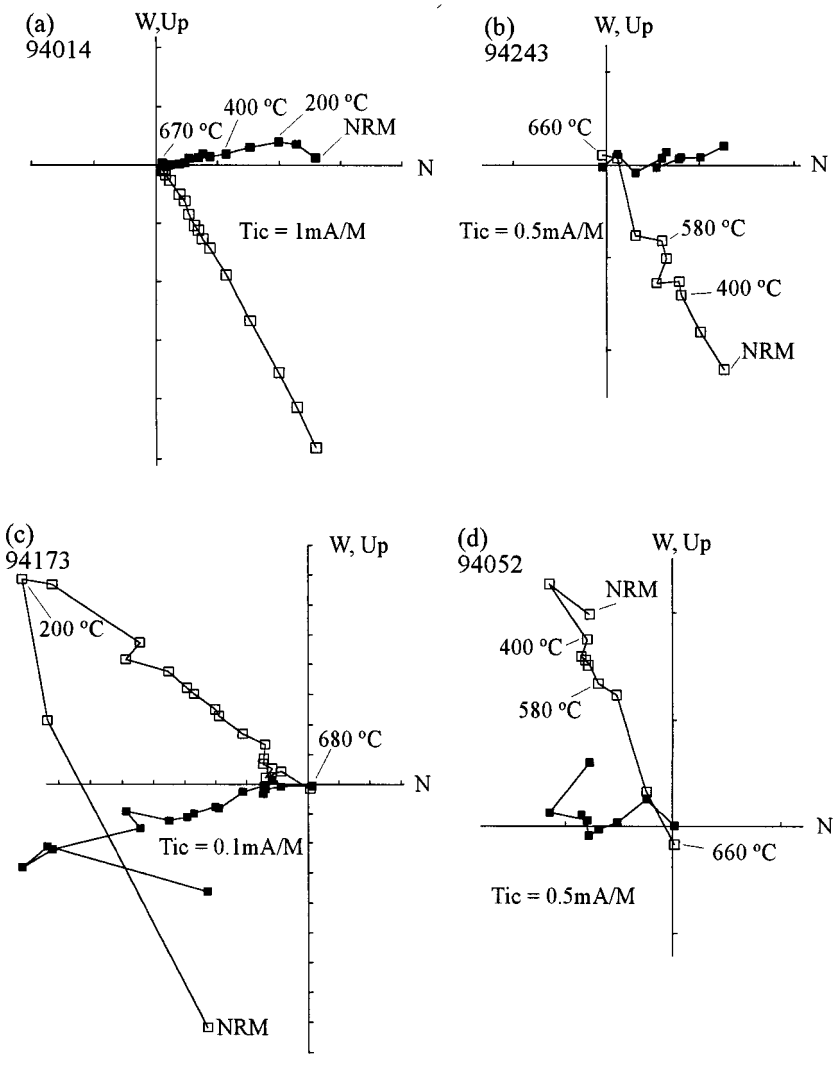


Figure 2. Representative Zijderveld (1967) diagrams of Wasatch Formation paleosol samples. (*a*) and (*b*) Thermal demagnetization of a present-day magnetization overprinting a normal polarity characteristic magnetization. (*c*) and (*d*) Thermal demagnetization of a present-day magnetization overprinting a reversed polarity characteristic magnetization. The open (solid) circles show vector endpoints in the vertical (horizontal plane).

day geomagnetic field direction. Its low unblocking temperature suggests a goethite or pigmentary hematite magnetic carrier associated with recent diagenesis and/or weathering. The characteristic component has unblocking temperatures up to 680°C and N, NW, and down, or S, SE, and up directions. This characteristic component is interpreted as primary based on antipodal directions and congruence with the expected Eocene pole for southwestern Wyoming (D = 349°, I = 61°; Diehl et al. 1983). This primary magnetization is carried by coarse-grained hematite formed during original pedogenesis. Previous results from the Willwood Formation (Eocene) in the Bighorn Basin of Wyoming (Clyde et al. 1994; Tauxe et al. 1994) and Black Prince Limestone (Pennsylvanian) in Arizona (Nick et al. 1991) show similar synsedimentary magnetization of hematite in paleosol horizons.

The grayish/green shales and limestones of the Green River Formation were generally characterized by weak NRM intensities (typically <1 mA/M) and more erratic demagnetization behavior (figure 3). Stable demagnetization was obtained from only 11 of the 69 Green River samples analyzed (16%). These stable samples were generally characterized by a weak overprint component with unblocking temperatures up to 300°C and a characteristic component with unblocking temperatures up to 550°C. The characteristic component is characterized by antipodal directions consistent with the expected Eocene pole for southwestern Wyoming and is interpreted as primary. Unblocking temperatures associated with the characteristic magnetization suggest magnetite carries the remanence. Some Green River Formation sites were also characterized by overlapping unblocking spectra of these two magnetic components, obscuring linear demagnetization trends. However, samples with

This content downloaded from 23.235.32.0 on Sat, 12 Dec 2015 08:41:09 AM
All use subject to JSTOR Terms and Conditions

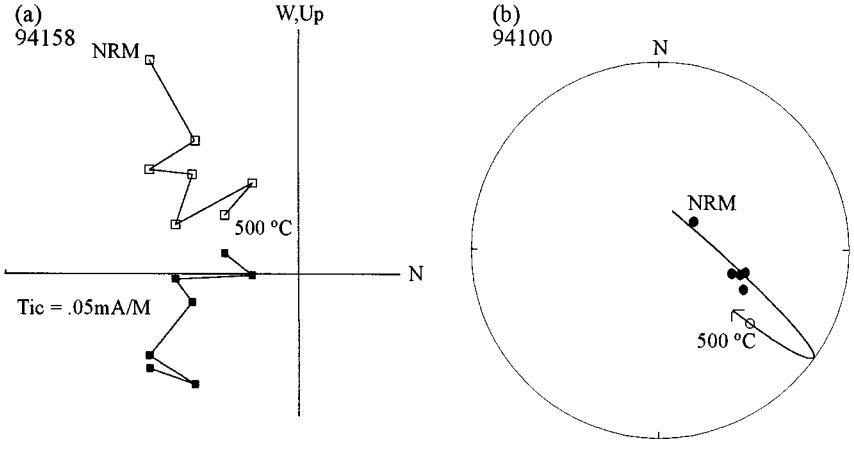


Figure 3. (*a*) Representative Zijderveld diagram of a Green River Formation limestone sample characterized by reversed polarity. Symbols as in figure 2. (*b*) Equal area projection showing a great circle fit to directions during progressive thermal demagnetization of a Green River Formation sample. The directions move from the present-day field direction toward a reversed polarity Eocene direction. Open (closed) symbols lie on the upper (lower) hemisphere of the projection.

remanence directions that followed a great circle path from the north and down overprint direction toward a reverse characteristic direction were used to augment our magnetostratigraphic interpretation (figure 3*b*).

A total of 129 characteristic directions were obtained from the original 312 samples collected. Most of these characteristic directions (118) came from paleosol samples. The average maximum angular deviation (MAD) of the least squares fit through the primary component of magnetization for stable samples is 12.0°. Normal and reversed samples appear antipodal and distinct from the present-day field direction (figure 4*a* and 4*b*). The reversed directions are more scattered than the normal directions because of incomplete removal of the present-day field. Grouping of sample directions into 25 alpha sites also shows well-clustered normal and antipodal reversed directions (figure 4*c*). Alpha sites are characterized by three or more samples of the same polarity and a mean kappa of 642.1. These sites pass the reversal test at the 95% confidence limit (McFadden and Lowes 1981) and have characteristic directions very consistent with the expected Eocene pole for southwestern Wyoming (Diehl et al. 1983).

A plot of the magnetic polarity results against pertinent stratigraphic information reveals six polarity zones (figure 5). The lowest 25 m of the section are characterized by normal polarity (zone A+), the next 230 m by dominantly reversed polarity (zone B−), and the next 100 m by alternating polarity (normal [zone C+], reverse [zone D−], normal [zone E+], and reverse [zone F−]). Polarity transitions do not coincide with lithological boundaries or shifts in magnetic susceptibility, suggesting the polarity record here is not affected by rock type or magnetic mineralogy. The Wasatchian/Bridgerian NALMA boundary falls within the long, reversed zone (zone B−) in the middle of the study section.

Correlation to the Geomagnetic Polarity Time Scale

A variety of previous magnetostratigraphic and radioisotopic studies from Laramide basin settings provide constraints on the correlation of the record studied here. The following is a list of these constraints. In some cases the results are poorly documented, making their reliability difficult to assess.

1) Magnetostratigraphic results from the Bighorn Basin indicate that the early through late Wasatchian NALMA correlates to Chron C24 (Butler et. al. 1981; Clyde et al. 1994; Tauxe et al. 1994).

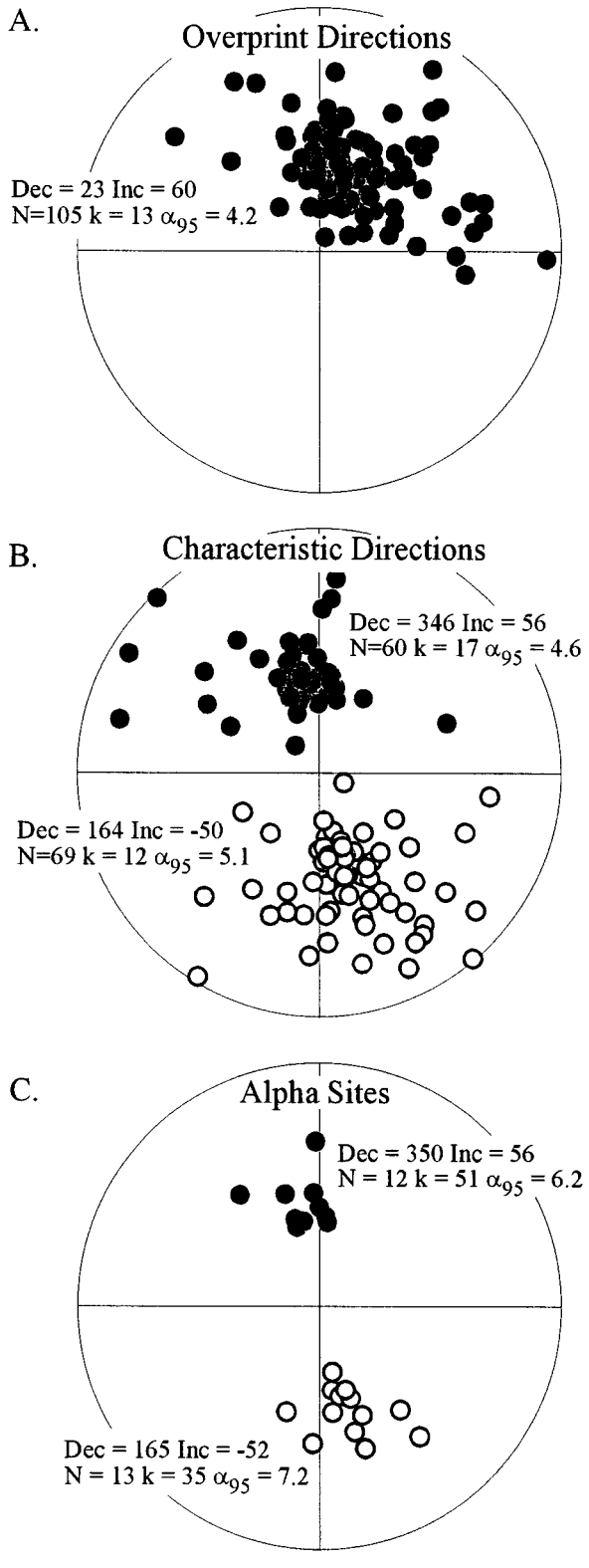
2) Magnetostratigraphic results from the East Fork Basin constrain an early Bridgerian assemblage to lie within Chron C21 or C22 (Flynn 1986; See also McCarroll et al. 1996 and Walsh 1996).

3) An Ar^{40}/Ar^{39} radioisotopic age of 52.8 ± 0.3 Ma from the central Bighorn Basin constrains the early Lostcabinian "subage" (late Wasatchian) to lie within the latter part of Chron C24n (Wing et al. 1991).

4) Magnetostratigraphic and radioisotopic results from the southeastern Absaroka Range support the correlation of early Bridgerian assemblages to Chron C22 (Sundell et al. 1984). No analytical information is provided in this study making the reliability of these results difficult to evaluate.

5) An Ar^{40}/Ar^{39} radiometric age of 47.3 ± 0.05 Ma for the Henry's Fork Tuff in the Green River Basin (reported in Prothero and Emry 1996, p. 664) correlates to the middle Bridgerian NALMA and lies approximately 400 m stratigraphically above the top of the section studied here (Evanoff pers.

This content downloaded from 23.235.32.0 on Sat, 12 Dec 2015 08:41:09 AM
All use subject to JSTOR Terms and Conditions



comm.). No analytical results were reported for this age, making it of uncertain reliability.

6) A variety of other K-Ar radiometric ages that date late Lostcabinian and early Bridgerian assemblages place the Wasatchian/Bridgerian NALMA boundary at approximately 50–51 Ma (Krishtalka et al. 1987, p. 91–95). However, the analytical reliability of most of these results is difficult to evaluate (e.g., Evernden et al. 1964; MacGinitie et al. 1974; Love et al. 1976), and the biostratigraphic control is often indirect (Mauger 1977).

Given these constraints, there are only two potential correlations of this Green River Basin record to the GPTS (figure 6; Cande and Kent 1995). The first ("Correlation 1"), correlates zone A+ to Chron C24n.1n, zone B− to C23r, zone C+ to C23n.2n, zone D− to C23n.1r, zone E+ to C23n.1n, and zone F− to C22r. The second ("Correlation 2"), correlates zone A+ to C23.1n, zone B− to C22r, zone C+ through E+ to C22n, and zone F− to C21r.

Correlation 1 fits the GPTS reversal pattern better than Correlation 2 but requires a substantial shift in sedimentation rate between zone B- and zone C+ (figure 6). Correlation 2 includes a short reversal within Chron C22n that is not documented in the GPTS record derived from marine surveys, but agrees with traditional placement of the Wasatchian/Bridgerian boundary between 50 and 51 Ma. Correlation 2 also indicates more uniform sedimentation rates through the section (figure 6).

Age of the Wasatchian/Bridgerian NALMA Boundary

The two correlations presented here pose significantly different estimates for the timing of the Wasatchian/Bridgerian NALMA boundary (figure 7). However, because of the constraints imposed by previous magnetostratigraphic and radioisotopic

Figure 4. Equal area projections of (*a*) overprint directions, (*b*) characteristic directions, and (*c*) alpha site directions. The open (closed) symbols represent sample directions on the upper (lower) hemisphere of the projection. Fisher (1953) statistics are shown within each projection. The overprint component of magnetization is characterized by north, north-east and down directions. The characteristic component of magnetization is characterized by N, NW, and down, or S, SE, and up directions. Alpha sites pass the reversal test at the 95% confidence limit (McFadden and Lowes 1981), and the observed mean declination/inclination, when reversed sites are inverted, is 347/54 which is very close to the expected Eocene direction of 349/61 (Diehl et al. 1983).

This content downloaded from 23.235.32.0 on Sat, 12 Dec 2015 08:41:09 AM
All use subject to JSTOR Terms and Conditions

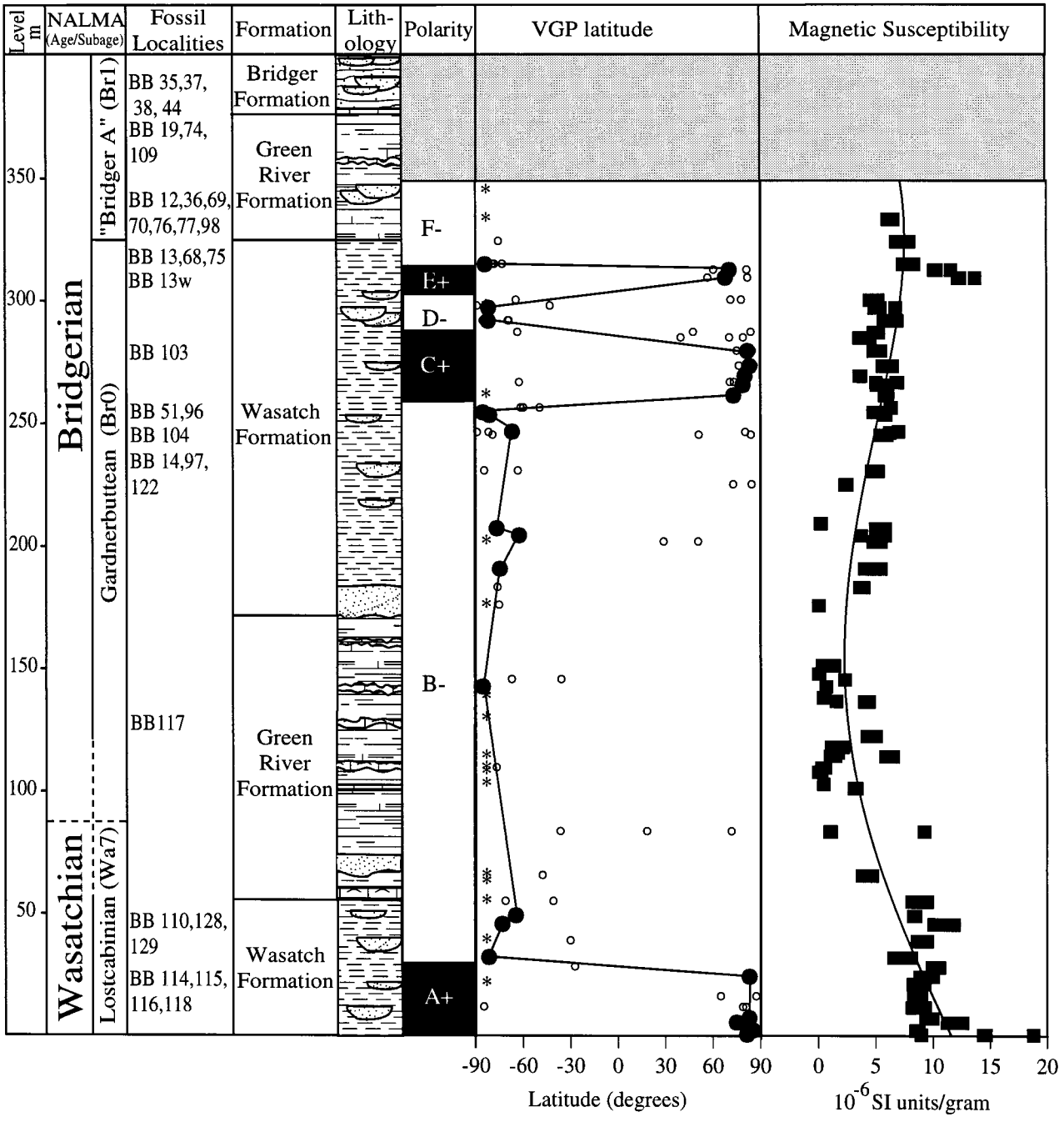


Figure 5. VGP latitude and magnetic susceptibility plotted against biostratigraphic and lithostratigraphic information from the study section. Solid circles represent alpha sites, open circles represent individual samples not included in alpha sites, and asterisks represent samples that follow a great circle path toward a reverse direction during demagnetization. A total of six polarity zones are identified in this record (zones A+ through F−). Notice that polarity reversals do not coincide with shifts in lithology or magnetic susceptibility, suggesting that this magnetostratigraphic record is not affected by rock type or magnetic mineralogy. Taxonomic lists for localities are given in tables 1 and 2.

This content downloaded from 23.235.32.0 on Sat, 12 Dec 2015 08:41:09 AM
All use subject to JSTOR Terms and Conditions

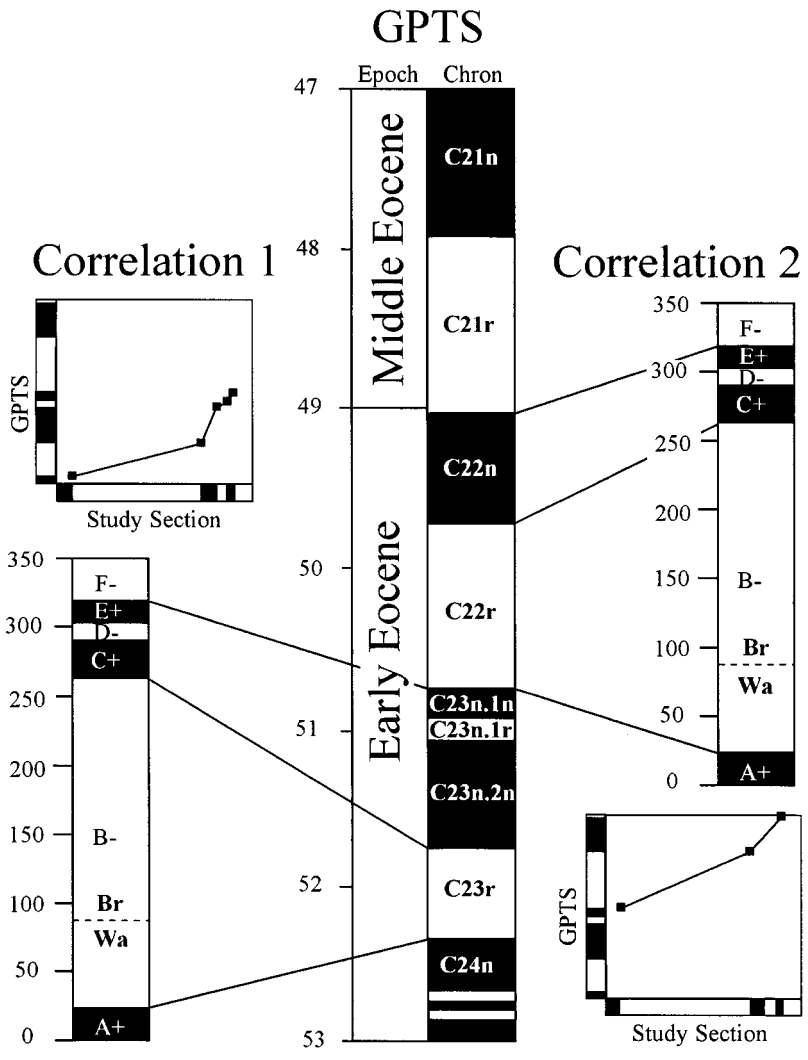


Figure 6. Two potential correlations of the study section to the GPTS (Geomagnetic Polarity Time Scale; Cande and Kent 1995; Berggren et al. 1995) given relevant radiometric and magnetostratigraphic constraints (see text). Correlation 1 places the Wasatchian/Bridgerian NALMA boundary (shown with a dashed line on polarity records) within Chron C23R and requires a significant change in sedimentation rates within the section. Correlation 2 places the Wasatchian/Bridgerian NALMA boundary within Chron C22r and requires only minor changes in sedimentation rates. Inset graphic correlations show relative changes in sedimentation rates given the corresponding correlation. Wa—Wasatchian NALMA; Br—Bridgerian NALMA.

results, these represent the only two correlations possible.

Correlation 1 is not consistent with previous estimates for the timing of this part of the NALMA in that it places the Wasatchian/Bridgerian boundary at around 52 Ma, approximately 1–2 myr older than estimates based on the K-Ar ages reviewed in Krishtalka et al. (1987). This correlation also implies a significant decrease in sedimentation rate during zone C+. Although the Green River Formation and Wasatch Formation may indeed be characterized by different sedimentation rates, there is no shift in lithology or magnetic susceptibility in zone C+ that indicates a dramatic decrease in sediment accumulation during this specific interval. This correlation would shorten the Wasatchian to 3.5 myr in duration and lengthen the Bridgerian to approximately 5 myr in duration.

We tentatively favor Correlation 2 since it agrees well with conventional estimates for the age of the Wasatchian/Bridgerian boundary (~50.5 Ma) and implies more uniform rates of sediment accumulation. Although this correlation is burdened by the presence of a reversal in the local section that is not recorded in the marine polarity record, short-term cryptochrons are poorly resolved in this part of the marine record (Cande and Kent 1992, p. 13,929). In fact, close inspection of two magnetic profiles from the Indian Ocean (GAL03 and MDF16; Cande and Kent 1992, fig. 23) shows potential reversals within Chron C22. If Correlation 2 is correct, then the duration of the Wasatchian and Bridgerian NALMAs would change little from previous estimates—approximately 5 and 3.5 myr respectively.

Both correlations put the Wasatchian/Bridgerian boundary in the early Eocene based on the new time scale of Berggren et al. (1995). This differs from many previous studies that have considered the Bridgerian to be entirely middle Eocene (e.g. Wood

This content downloaded from 23.235.32.0 on Sat, 12 Dec 2015 08:41:09 AM
All use subject to JSTOR Terms and Conditions

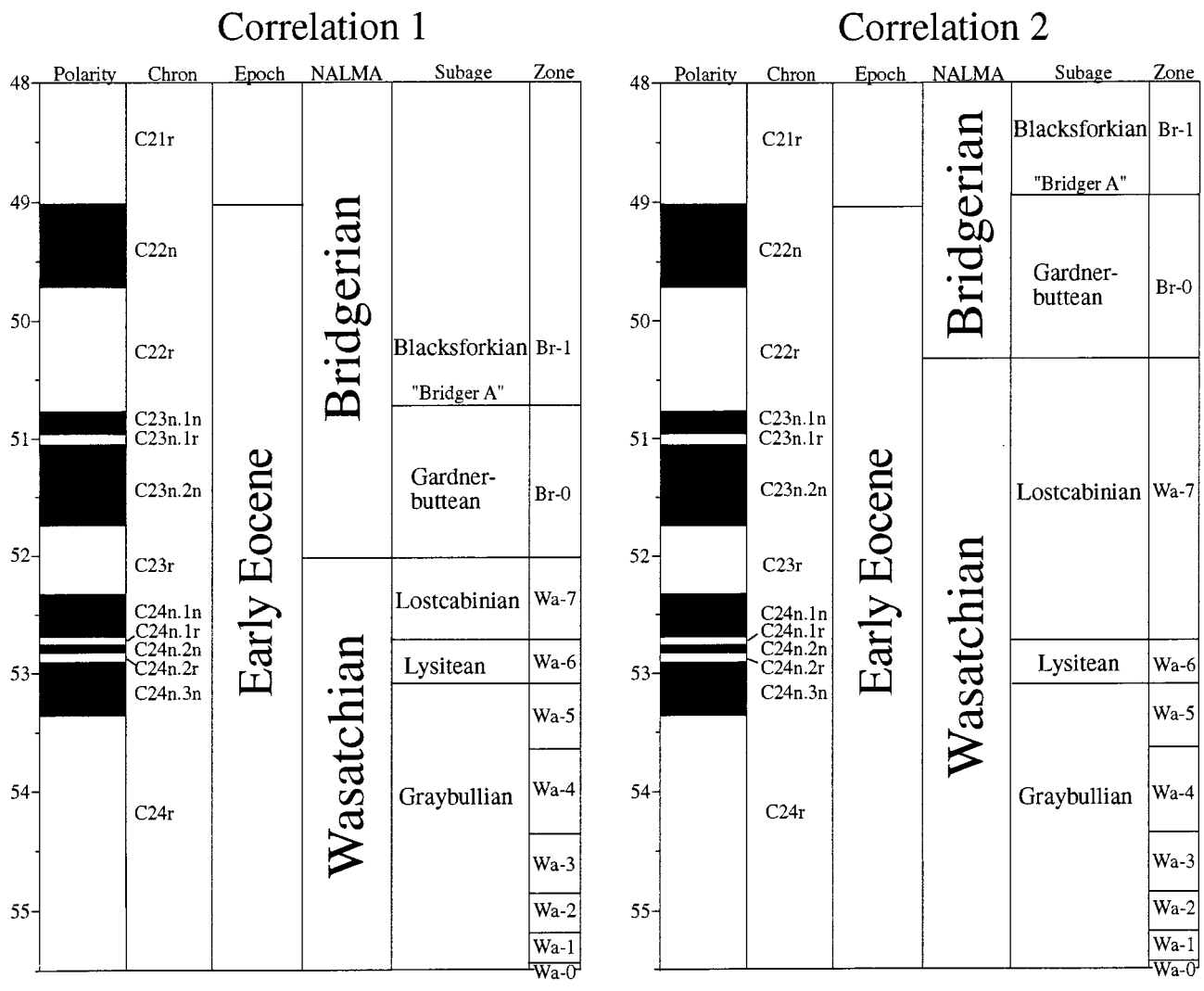


Figure 7. Chronological implications for the two potential correlations. Correlation 1 places the Wasatchian/Bridgerian boundary at approximately 52 Ma and Correlation 2 places the boundary at approximately 50.5 Ma. Correlation 2 is tentatively favored because it requires no abrupt change in sediment accumulation rate and is more congruent with conventional estimates for the age of this boundary. The time scale is based on Cande and Kent (1995) and Berggren et al. (1995). Chronology of early to middle Wasatchian is based on Clyde et al. (1994) and Clyde (1997).

et al. 1941; Berggren et al. 1985; Prothero 1995, p. 305). No matter which correlation is correct, we can now state with confidence that the Wasatchian/Bridgerian NALMA boundary predates the Ypresian/Lutetian age boundary in Europe.

Although we have established that the Wasatchian/Bridgerian NALMA boundary falls either within Chron C23r or Chron C22r, it will require continued radioisotopic and/or magnetostratigraphic work to constrain further the correlation of this land mammal age boundary to the GPTS. Magnetostratigraphic analysis of a continuous stratigraphic section that spans the middle Wasatchian (Lysitean "subage") through early Bridgerian would be the most direct way to resolve this question. Timing of the Lostcabinian "subage" is most in question since Correlation 1 implies that it lasts some 0.5 myr and represents only about one-seventh of Wasatchian time, whereas Correlation 2 implies that it lasts over 2.5 myr and represents more than one-half of Wasatchian time (given the correlation of the earliest Lostcabinian to Chron 24n.1 in the Bighorn Basin: Clyde et al. 1994; Tauxe et al. 1994). A high precision Ar^{40}/Ar^{39} age for the late Lostcabinian would also help resolve this question, although finding dateable rocks during this interval has proven difficult.

This content downloaded from 23.235.32.0 on Sat, 12 Dec 2015 08:41:09 AM
All use subject to JSTOR Terms and Conditions

Magnetic Susceptibility

Bulk susceptibility of samples was measured to explore the potential relationship between magnetic mineralogy and depositional environment in this fluvial/lacustrine setting. Susceptibility, which is the magnetization of rocks in response to an applied field, is indicative of magnetic and paramagnetic mineralogy. Since subtle changes in magnetic mineralogy often reflect subtle changes in depositional or diagenetic conditions, magnetic susceptibility has been used as a proxy for various depositional and environmental variables (e.g., Maher and Thompson 1992; Heller et al. 1993; Maher et al. 1994).

In our study area, Green River Formation shales and limestones deposited during high lake levels are generally characterized by low susceptibility values, and Wasatch Formation paleosols deposited during relatively low lake levels are characterized by high susceptibility values (figure 5). Pedogenesis seems to have elevated susceptibility values in these fluviolacustrine sediments, probably from pedogenic growth of paramagnetic clay minerals. A weak relationship between pedogenesis and magnetic susceptibility has also been observed in Willwood Formation paleosols in the Bighorn Basin (Clyde and Stamatakos 1994).

The most common method for ''measuring'' paleosol maturity (i.e., degree of pedogenesis) is by visual inspection (e.g., Bown and Krause 1987). This is a somewhat subjective method that varies from worker to worker. If the relationship between magnetic susceptibility and pedogenesis recognized here is found to be a general phenomenon upon further study, magnetic susceptibility may provide a quantitative proxy for paleosol maturity that could replace the visual coding system now in wide use.

Conclusion

Paleomagnetic analysis of a 350-m section of fluviolacustrine sediments (Wasatch and Green River formations) from the western Green River Basin of Wyoming provides a magnetobiostratigraphic framework to correlate the late Wasatchian through early Bridgerian North American Land Mammal Ages to the Geomagnetic Polarity Time Scale. Thermal demagnetization of samples resulted in a nearly antipodal distribution of remanent directions consistent with the expected Eocene direction for this area. Given present radiometric and magnetostratigraphic constraints, two potential correlations of the resulting polarity record to the GPTS exist. Correlation 1 places the Wasatchian/Bridgerian NALMA boundary within Chron C23r at approximately 52 Ma, and Correlation 2 places the boundary within Chron C22r at approximately 50.5 Ma. Correlation 2 is tentatively favored since it assumes more uniform rates of sediment accumulation and agrees more closely with previous estimates for the age of this boundary.

ACKNOWLEDGMENTS

We thank J. Alroy, J. Bloch, J. Flynn, P. Gingerich, P. Robinson, and R. Van der Voo for helpful discussion. We also thank R. Ralls for his help in the field. Three anonymous reviewers significantly improved the manuscript. This research was funded by the Geological Society of America and the National Geographic Society.

REFERENCES CITED

Berggren, W. A.; Kent, D. V.; and Flynn, J. J., 1985, Paleogene geochronology and chronostratigraphy, *in* Snelling, N. J., ed., The Chronology of the Geological Record: Geol. Soc. Lond. Mem.10, p. 141–195.

———; ———; Swisher, C. C.; and Aubry, M-P., 1995, A revised Cenozoic geochronology and chronostratigraphy, *in* Berggren, W. A.; Kent, D. V.; Aubry, M-P.; and Hardenbol, J., eds., Geochronology, Time Scales and Global Stratigraphic Correlation: SEPM Spec. Pub. 54, p. 129–212.

Bown, T. M., and Kraus, M. J., 1987, Integration of channel and floodplain suites, I. Developmental sequence and lateral relations of alluvial paleosols: Jour. Sed. Pet., v. 57, p. 587–601.

Butler, R. F.; Gingerich, P. D.; and Lindsay, E. H., 1981, Magnetic polarity stratigraphy and biostratigraphy of Paleocene and lower Eocene continental deposits, Clarks Fork Basin, Wyoming: Jour. Geology, v. 89, p. 299–316.

Cande, S. C., and Kent, D. V., 1992, A new geomagnetic polarity time scale for the Late Cretaceous and Cenozoic: Jour. Geophys. Res., v. 97, p. 647–657.

———, and ———, 1995, Revised calibration of the geomagnetic polarity time scale for the Late Cretaceous and Cenozoic: Jour. Geophys. Res., v. 100, p. 6093–6095.

Clyde, W. C., 1997, Stratigraphy and mammalian paleontology of the McCullough Peaks, northern Bighorn Basin, Wyoming: Implications for biochronology, basin development, and community reorganization across the Paleocene-Eocene boundary: Unpub. Ph.D. Dissertation, University of Michigan, Ann Arbor.

———, and Stamatakos, J., 1994, A magneto-environmental record from the Bighorn Basin of Wyoming:

This content downloaded from 23.235.32.0 on Sat, 12 Dec 2015 08:41:09 AM
All use subject to JSTOR Terms and Conditions

Trans. Am. Geophys. Union, Prog. with Abs., p. 187.

———; ———; and Gingerich, P. D., 1994, Chronology of the Wasatchian Land-Mammal Age (early Eocene): Magnetostratigraphic results from the McCullough Peaks section, northern Bighorn Basin, Wyoming: Jour. Geology, v. 102, p. 367–377.

Collinson, M. E., and Hooker, J. J., 1987, Vegetational and mammalian faunal changes in the early Tertiary of southern England, *in* Friis, E. M.; Chaloner, W. G.; and Crane, P. R., eds., The Origins of Angiosperms and their Biological Consequences: Cambridge University Press, p. 259–304.

Diehl, J. F.; Beck, M. E., Jr.; Beske-Diehl, S.; Jacobson, D.; and Hearn, B. C., Jr., 1983, Paleomagnetism of the late Cretaceous-early Tertiary north-central Montana alkalic province: Jour. Geophys. Res., v. 88, p. 10, 593–10,609.

Evernden, J. R.; Savage, D. E.; Curtis, G. H.; and James, G. T., 1964, Potassium-argon dates and the Cenozoic mammalian chronology of North America: Am. Jour. Sci., v. 262, p. 145–198.

Fisher, R. A., 1953, Dispersion on a sphere: Proc. Royal Soc. (London), v. A217, p. 295–305.

Flynn, J. J., 1986, Correlation and geochronology of middle Eocene strata from the western United States: Palaeogeog., Palaeoclim., Palaeoecol., v. 55, p. 335–406.

Gunnell, G. F., 1997, Mammalian faunas from the lower Bridger Formation (lithostratigraphic horizon A, biostratigraphic zone Br-1), early middle Eocene, southern Green River Basin, Wyoming: Contrib. Mus. Paleont. Univ. Michigan, in press.

———, 1997, Wasatchian-Bridgerian (Eocene) paleoecology of the western interior of North America: Changing paleoenvironments and taxonomic composition of omomyid (Tarsiiformes) primates: Jour. Human Evolution, v. 32, p. 105–132.

———, and Bartels, W. S., 1994, Early Bridgerian (middle Eocene) vertebrate paleontology and paleoecology of the southern Green River Basin, Wyoming: Univ. Wyoming Contrib. Geol., v. 30, p. 57–70.

Heller, F.; Shen, C. D.; Beer, J.; Liu, X. M.; Liu, T. S.; Bronger, A.; Suter, M.; and Bonani, G., 1993, Quantitative estimates of pedogenic ferromagnetic mineral formation in Chinese loess and palaeoclimatic implications: Earth Planet. Sci. Lett., v. 114, p. 385–390.

Hutchison, J. H., 1982, Turtle, crocodilian, and champsosaur diversity changes in the Cenozoic of the north-central region of western United States: Palaeogeog., Palaeoclim., Palaeoecol., v. 37, p. 149–164.

Kirschvink, J. L., 1980, The least-squares line and plane and the analysis of paleomagnetic data: Geophys. Jour. Royal Astron. Soc., v. 62, p. 743–746.

Krishtalka, L.; West, R. M.; Black, C. G.; Dawson, M. R.; Flynn, J. J.; Turnbull, W. D.; Stucky, R. K.; McKenna, M. C.; Bown, T. M.; Golz, D. J.; and Lilligraven, J. A., 1987, Eocene (Wasatchian through Duchesnean) biochronology of North America, *in* Woodburne, M. O., ed., The Cenozoic History of Mammals in North America: Berkeley, University of California Press, p. 77–117.

Love, J. D., and Christiansen, A. C., 1985, Geologic map of Wyoming: Dept. of the Interior, U.S. Geol. Survey Map, scale 1:500,000.

———; McKenna, M. C.; and Dawson, M. R., 1976, Eocene, Oligocene, and Miocene rocks and vertebrate fossils at the Emerald Lake locality, three miles south of Yellowstone National Park, Wyoming: U.S. Geol. Survey Prof. Paper 932-A.

MacGinitie, H. D.; Leopold, E. B.; and Rohrer, W. L., 1974, An early middle Eocene flora from the Yellowstone-Absaroka volcanic province, northwestern Wind River Basin, Wyoming: Univ. Calif. Pub. Geol. Sci., v. 108, p. 1–103.

Maher, B. A.; Thompson, R.; and Zhou, L. P., 1994, Spatial and temporal reconstructions of changes in the Asian palaeomonsoon: A new mineral magnetic approach: Earth Planet Sci. Lett., v. 125, p. 461–471.

———, and Thompson, R., 1992, Paleoclimatic significance of the mineral magnetic record of the Chinese loess and paleosols: Quat. Res., v. 37, p. 155–170.

Markwick, P., 1994, ''Equability,'' continentality, and Tertiary ''climate'': The crocodilian perspective: Geology, v. 22, p. 613–616.

Mauger, R. L., 1977, K-Ar ages of biotites from tuffs in Eocene rocks of the Green River, Washakie, and Uinta basins, Utah, Wyoming, and Colorado: Univ. Wyoming Contrib. Geol., v. 15, p. 17–41.

McCarroll, S. M.; Flynn, J. J.; and Turnbull, W. D., 1996, Biostratigraphy and magnetostratigraphy of the Bridgerian-Uintan Washakie Formation, Washakie Basin, Wyoming, *in* Prothero, D. R., and Emry, R. J., eds., The Terrestrial Eocene-Oligocene Transition in North America: Cambridge, Cambridge University Press, p. 25–39.

McFadden, P. L., and Lowes, F. J., 1981, The discrimination of mean directions drawn from Fisher distributions: Geophys. Jour. Royal Astron. Soc., v. 67, p. 19–33.

M′Gonigle, J. W., and Dover, J. H., 1992, Geologic map of the Kemmerer 30′ × 60′ quadrangle, Lincoln, Uinta, and Sweetwater Counties, Wyoming: Dept. of the Interior, U.S. Geol. Survey Map, scale 1:100,000.

Nick, K.; Xia, K.; and Elmore, R. D., 1991, Paleomagnetic and petrographic evidence for early magnetization in successive Terra Rose paleosols, Lower Pennsylvanian Black Prince Limestone, Arizona: Jour. Geophys. Res., v. 96, p. 9873–9885.

Prothero, D. R., 1995, North American land mammal ''ages'', *in* Berggren, W. A.; Kent, D. V.; Aubry, M-P.; and Hardenbol, J., eds., Geochronology, Time Scales, and Global Stratigraphic Correlation: SEPM Spec. Pub. 54, p. 295–315.

———, and Emry, R. J., eds., 1996, The Terrestrial Eocene-Oligocene Transition in North America: Cambridge, Cambridge University Press, 688 p.

Robinson, P., 1966, Fossil Mammalia of the Huerfano Formation, Eocene, of Colorado: Bull. Peabody Mus. Nat. Hist., v. 21, p. 1–25.

This content downloaded from 23.235.32.0 on Sat, 12 Dec 2015 08:41:09 AM
All use subject to JSTOR Terms and Conditions

Stucky, R. K., 1984, The Wasatchian-Bridgerian land mammal age boundary (early to middle Eocene) in western North America: Ann. Carnegie Mus., v. 53, p. 347–382.

Sundell, K. A.; Shive, P. N.; and Eaton, J. G., 1984, Measured sections, magnetic polarity and biostratigaphy of the Eocene Wiggins, Tepee Trail, and Aycross formations within the southeastern Absaroka Range, Wyoming: Earth Sci. Bull., v. 17, p. 1–48.

Tauxe, L.; Gee, J.; Gallet, Y.; Pick, T.; and Bown, T., 1994, Magnetostratigraphy of the Willwood Formation, Bighorn Basin, Wyoming: New constraints on the location of the Paleocene/Eocene boundary: Earth Planet. Sci. Lett., v. 125, p. 159–172.

Walsh, S. L., 1996, Theoretical biochronology, the Bridgerian/Uintan boundary, and the ''Shoshonian subage'' of the Uintan, *in* Prothero, D. R., and Emry, R. J., eds., The Terrestrial Eocene-Oligocene Transition in North America: Cambridge, Cambridge University Press, p. 52–74.

Wing, S. L.; Bown, T. M.; and Obradovich, J. D., 1991, Early Eocene biotic and climatic change in interior western North America: Geology, v. 19, p. 1189–1192.

———, and Greenwood, D. R., 1993, Fossils and fossil climate: The case for equable continental interiors in the Eocene: Phil. Trans. Royal Soc. (London), v. 341, p. 243–252.

Wood, H. E., II; Chaney, R. W., Jr.; Clark, J.; Colbert, E. H.; Jepsen, G. L.; Reeside, J. B.; and Stock, C., 1941, Nomenclature and correlation of the North American continental Tertiary: Geol. Soc. America Bull., v. 52, p. 1–48.

Woodburne, M. O., and Swisher, C. C., 1995, Land mammal high-resolution geochronology, intercontinental overland dispersals, sea level, climate, and vicariance, *in* Berggren, W. A.; Kent, D. V.; Aubry, M.-P.; and Hardenbol, J., eds., Geochronology, Time Scales and Global Stratigraphic Correlation: SEPM Spec. Pub. 54, p. 335–364.

Zachos, J. C.; Stott, L. D.; and Lohmann, K. C., 1994, Evolution of early Cenozoic marine temperatures: Paleoceanography, v. 9, p. 353–387.

Zijderveld, J. D. A., 1967, A.C. demagnetization of rocks: analysis of results, *in* Collinson, D. W.; Creer, K. M.; and Runcorn, S. K., eds., Methods of Palaeomagnetism: Amsterdam, Elsevier, p. 254–286.

Zonneveld, J.-P., 1994, The Wasatchian-Bridgerian land mammal age boundary (early to middle Eocene) in the Desertion Point–Little Muddy area, southwestern Green River Basin, Wyoming: Unpub. M.S. Thesis, Michigan State University, East Lansing, 248 p.

This content downloaded from 23.235.32.0 on Sat, 12 Dec 2015 08:41:09 AM
All use subject to JSTOR Terms and Conditions

This content downloaded from 23.235.32.0 on Sat, 12 Dec 2015 08:41:09 AM
All use subject to JSTOR Terms and Conditions